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Brac University
Semester: Fall 2025
Course Code: CSE430
Digital Signal Processing
Faculty: EHQ

Assessment: *Quiz 2 + 3*
Duration: 40 minutes
Date: November 19, 2025
Full Marks : 20 + 20

■ Question 1 of 2 [20 marks]

A brief three channeled EEG signal is provided below. The units against each value is in μV .

Channel 1	3	5	7	k_x	5	9
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Channel 2	-13	-8	-3	3	k_x	7
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Channel 3	7	k_x	-9	-6	7	13
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Use $k_x = \text{mod}(\text{Student ID}, 7) + 1$ for this question.

For all operations use the truncated variants. You have to work out only one operation for all the questions.

- (a) [6 marks] Filter the signal using the kernel, $k_1 = [-1 \ 0 \ 1]$
- (b) [6 marks] Filter the signal using the kernel $k_2 = [-1 \ 2 \ -1]$
- (c) [8 marks] Find $\text{ReLU}(\sum_{i=1}^3 \text{Channel}_i \star k_2)$

(a) Reverse the kernel for convolution. Convolve each channel separately. Add all the channels to get a single channeled output.

(b) Same as (a). Do note that the kernel is symmetrical. Something that will come in handy in the next question.

(c) The expression given here is basically a filtering equation, except filtering using convolution, you have to use cross-correlation. At first, the expression is asking you to find the cross-correlation of each channel using the kernel in question (b) and then it is asking you to sum up the cross-correlation products of every channel pointwise. This is exactly what you did in (a) and (b).

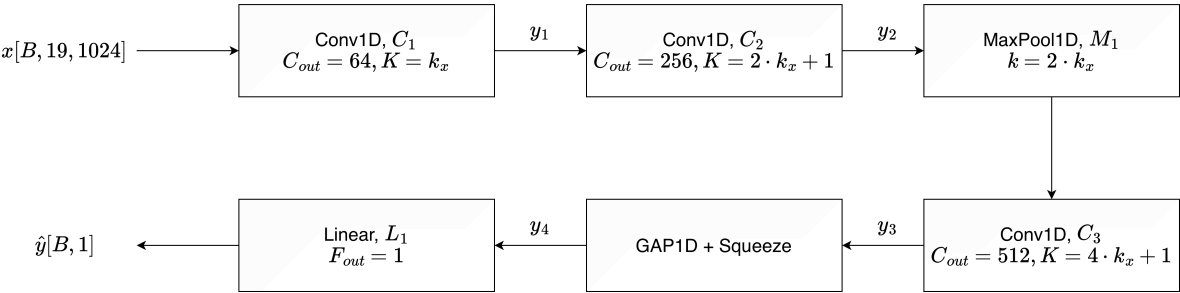
Now the trick is since the kernel is symmetric, the filtering result for (c) will be the same as the result in (b). So just take the answer of (b) and apply ReLU on it and you have successfully answered (c).

Do note that whenever you are using answers from a previous question, you have to mention why you are doing it and why it would work.

■ Question 2 of 2 [20 marks]

Provided below is an EEG classifier model to detect the presence of epileptic artifacts in a patient’s reading. Your task is to fill up the table provided below using information from the model. Assume the batch size to be 32 and bias = True for all applicable layers.

For this question use $k_x = 2 \cdot \text{mod}(\text{Student ID}, 4) + 3$ C_in is not given for any layers. You have to get the value of C_in from the input signal.
Assume bias = True for all relevant layers.



Layer	Parameter count	Output variable	Dimensions
C_1	$19 \cdot 64 \cdot k_x + 64$	y_1	$[32, 64, 1024 - k_x + 1]$
C_2	$64 \cdot 256 \cdot (2k_x + 1) + 256$	y_2	$[32, 256, 1025 - 3k_x]$
M_1	0	y_3	$[32, 512, (1025 - 3k_x) // 2 - 4k_x]$
C_3	$256 \cdot 512 \cdot (4k_x + 1) + 512$	y_4	$[32, 512]$
L_1	$512 \cdot 1 + 1$	$\hat{y}$	$[32, 1]$